Starters: From Scratch, Dried, and Revived

The 14-day method, what is happening in the jar, and a backup that comes back from a pinch

Sourdough Science

github.com/zrrbite/sourdough-science

What this deck covers

1. **The 14-day method**: Full Proof Baking's schedule in grams, and her rules
2. **What is happening in the jar**: the early rise is bacteria, the day-5 lull is the hand-over, what the settled culture is, and why a starter held at 1:2:2 fades
3. **Video, literature and this kitchen**: where the explanations differ; warm, not hot
4. **Drying a backup and reviving it**: a pinch, not a spoonful
5. **Open tests**: eleven experiments the note leaves open

Flags carried over from the notes: **hypothesis** **not yet tested** **not peer-reviewed** **loose analogue**.
Results from this kitchen are marked **own result**; everything else is cited at the foot of the slide.

Source: notes/starter-from-scratch.md and notes/starter-cheatsheet.md

The 14-day method

What you need: a 30% rye mix, a rubber band and 24°C

- **Flour mix, 30% rye:** 350 g bread flour + 150 g dark rye, whisked together. Every feed until day 12 uses it
- **A straight-sided jar** with a rubber band at the level of each fresh feed; a fresh jar on day 2, and whenever it gets messy
- **A scale reading to 0.1 g**, tared before each ingredient: the feeds go down to 5 g
- **24°C** (75°F), "warm room temperature", for all 14 days
- **Cheesecloth** and a rubber band until day 8, then a loosely fitted lid

Ratios are **starter : water : flour by weight**, so every feed is at 100% hydration.

In the kitchen: Rye for its nutrients and enzymes, bread flour for strength. Plain flour also works for the base.

Source: Kristen, Full Proof Baking (2019), from her on-screen title cards and narration

Two weeks in grams: 1:1:1 for a week, then the ratio climbs

Day	Morning, g	Evening, g	What you'll see
1	—	0 / 30 / 30	Mix only, rest 24 h
2	—	Stir, clean jar	Dry skin on top: fine
3	—	20 / 20 / 20	Risen: bacteria, not yeast yet
4	20 / 20 / 20	16 / 16 / 16	3–4×, runny, cheesy smell
5	**15 / 15 / 15**	**16 / 16 / 16**	**Almost no rise, vomit smell: normal**
6–7	15 / 15 / 15	15 / 15 / 15	1.2–2×, milder, first sour notes
8	15 / 15 / 15	15 / 15 / 15	~2×. Save discard, loose lid
9	15 / 15 / 15	8 / 16 / 16	2.5×
10	8 / 16 / 16	5 / 15 / 15	2.5×, falls before the feed
11	5 / 20 / 20	5 / 25 / 25	2.5×
12	5 / 25 / 25	5 / 20 / 20	Evening: switch to 10% rye mix
13	5 / 25 / 25	5 / 25 / 25	2.5×
14	8 / 16 / 16 ×2	10 / 50 / 50	Feeds 7 h apart; three a day now

Source: Full Proof Baking (2019), as tabulated in the cheat sheet. Feeds in grams of starter / water / flour: two a day from day 4, three from day 14

Feed at peak; it is done when it triples in 4–6 h at 27°C

Feeding

- Fallen before the feed? Next feed one ratio up: 1:2:2 → 1:3:3 → 1:4:4 → 1:5:5
- Fewer than three feeds a day: higher ratios, up to 1:10:10
- Day 12: 10% rye mix = leftover 30% mix + twice its weight in bread flour, for a “higher peak, hold peak longer”
- **Discard** if pink, orange or fuzzy

Done

- Triples in 4–6 h on 1:2:2 at **27°C**, not room temperature
- No 40°C+ builds until done. Fridge: 2–4 weeks unfed

Video, day 14	2× at	Peak
1:5:5	5 h	2.9× at 7.5 h
1:2:2	4 h	3× at 7 h

Not there yet: “still pretty young”

In the kitchen: Her closing card: “Feed at peak! Monitor your starter! Keep it warm!”

Source: Full Proof Baking (2019); the day-14 time-lapse, in an oven with the light on at 27°C (80°F)

What is happening in the jar

The first two weeks are a succession of microbes

	Days 1–4 the first wave	Day 5 the lull	Days 6–9 the hand-over	Days 9–14 building strength
In the jar	*Leuconostoc*, *Lactococcus*, *Enterococcus*, *Enterobacter* from the flour	Acid stops the first wave; the sourdough organisms are not yet in numbers	*Lactobacillus*, *Pediococcus*, *Weissella* take over; yeast the day after pH 3.5	Community largely settled; what remains is matching food to appetite
She saw	3–4× rise, unpleasant smells	“No significant growth!”, the worst smell	1.2× → 2×; “bit of sour” on day 7	Steady 2.5×; the ratio climbs

In the kitchen: The early rise that dies and the smell put people off, but they are an ordinary, well-documented succession.

Source: The two lab studies, both refreshed daily: Van der Meulen et al. (2007) at 30°C; Ercolini et al. (2013) 5 h a day at 25°C, 10°C between. Also Wink (2008a, b); Full Proof Baking (2019)

Days 1–4: the early rise is bacteria, not yeast

- Flour brings mostly *Proteobacteria*: *Pantoea*, *Pseudomonas*, *Enterobacter*. Within a day all but the *Enterobacteriaceae* are suppressed and lactic acid bacteria (LAB) take over
- The first LAB are not sourdough organisms: *Leuconostoc*, *Lactococcus*, *Enterococcus*. At 30°C they fall below detection within a day or two
- Rye is the exception: it arrives with *Weissella* already dominant
- Wink had a day-two starter analysed by two labs. The gas came from *Leuconostoc citreum*, with a little *Enterobacter cloacae*; neither lab found lactobacilli or yeast **not peer-reviewed**
- She links *Enterobacter*, *Aerococcus* and *Leuconostoc* to the bad smell

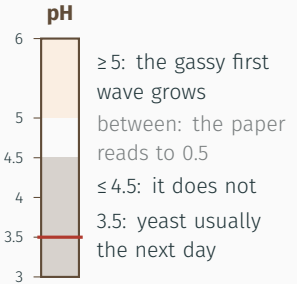
In the kitchen: The 3–4× rise and the smell on day 4 are this first wave. Keep to the schedule.

Source: Ercolini et al. (2013); Van der Meulen et al. (2007); Wink (2008b), not peer-reviewed

Day 5: the lull is acid shutting down the first wave

- As acid builds, the first wave stops before the sourdough organisms have numbers: “No significant growth!”
- Wink’s pH paper: gassy bacteria grew at pH ≥ 5, not ≤ 4.5; yeast usually came the day after pH 3.5

  careful amateur work; no replicate counts
- **hypothesis** (hers): *L. citreum*’s bacteriocins hold back the lactobacilli, so a bigger early rise means a longer lull
- Lab data agree: at 30°C, pH below 4.0 after day 2; yeast at 10^7 CFU/g after 4–5 days (one dough in four: 8)



In the kitchen: Almost no rise and a vomit smell on day 5 is normal. A pH strip at each feed, days 3–7, tests Wink’s thresholds.

Source: Wink (2008a, b), not peer-reviewed; Van der Meulen et al. (2007)

Days 6–9: the sourdough bacteria take over, then little changes

Sourdough-specific LAB, such as *Lactobacillus*, *Pediococcus* and *Weissella*, replace the first wave.

Where	Conditions	When it settles
Van der Meulen et al. (2007)	30°C, refreshed daily	Well-adapted strains dominate from about day 5
Ercolini et al. (2013)	25°C, 5 h a day	Mature in 5–7 days; lactobacilli prominent by day 10
The video	24°C	“Bit of sour” on day 7: about a day behind 30°C

In the kitchen: The first sour smell, with the rise back to 2×, is the hand-over: sourdough bacteria replacing the first wave.

Source: Van der Meulen et al. (2007); Ercolini et al. (2013); Full Proof Baking (2019)

Days 9–14: raising the ratio is about timing, not species

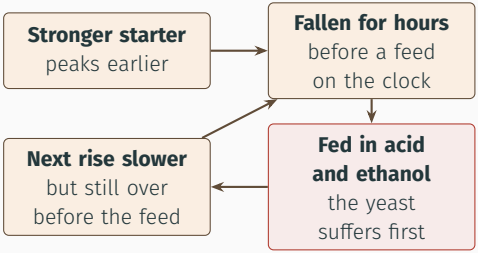
- By day 9 the community has largely settled. What remains is matching food to appetite
- A 1:2:2 feed that peaks and collapses within 12 h ran out of food well before the next feed, and then sits in its own acid
- A higher ratio dilutes the starter, so it takes longer to peak, and dilutes the acid it carried over: the same lever as chapter 11's recovery at 1:10:10
- "Feed at peak" is **a practitioner's rule, not a measured result**
- The plausible mechanism is chapter 11's compounding stressors: past peak the food is spent and acid and ethanol are at their highest, so a starter always fed after collapse is always fed at its most stressed

In the kitchen: If it peaks well before the next feed is due, raise the ratio. That is the whole of days 9–13.

Source: Full Proof Baking (2019); chapter 11, *Compounding Stressors* and *Recovery Protocol*

Hold the ratio while it gets faster, and it fades own result

- Held at 1:2:2 to 1:3:3 twice a day through its second week, often fed after it had fallen
- After the lull: several strong rises, then markedly slower within days
- As it strengthened its peak came earlier, so more of each 12 h was spent collapsed in acid and ethanol, where the yeast suffers first
- Nothing was caught or killed: the species were settled, only the yeast's share fell. Dilution should bring it back (open test 9)



A fixed ratio on a fixed clock, October 2026

In the kitchen: Held to the clock, the better it gets, the harder each feed punishes it.

Source: Own run, October 2026; Gänzle et al. (1998); Carbonetto et al. (2020); chapter 11, *Compounding Stressors*

Set the ratio by the interval, and read the top of the jar

The ratio sets where the peak lands. Pick the interval the day allows, then the ratio that puts the peak at it.

A starter that is getting strong, at 24–27° C

Feeds a day	Ratio
Three	1:2:2
Two	1:4:4 to 1:5:5
One	1:10:10

At feed time, the top of the jar decides the next ratio

- **Domed and still climbing:** not yet. Wait
- **Level, the first wrinkles or a dimple:** at peak. Feed now
- **Sunken, liquid on top, boozy:** it fell hours ago. Feed, and go one ratio up

Cooler or stiffer slows it too, but both change the regime the culture adapts to; the ratio does not. Practitioners' signs, not measured ones.

Starting points, not settings: her day 14 ran 1:2:2 three times a day and 1:5:5 overnight. A young starter may take as long on 1:5:5 as on 1:2:2 (7.5 h against 7 h), and a cool kitchen stretches every row.

In the kitchen: Held at 1:2:2 twice a day into week two, a good starter fades.

Source: Full Proof Baking (2019), *Her rules* and the day-14 time-lapse; own run, October 2026

Planning the next feed hypothesis

Estimate the activity first: time to peak at the current ratio and temperature, from the last two or three feeds. If each peaked earlier than the one before, go up a step now.

- **Ratio buys hours.** The whole ladder, 1:2:2 to 1:10:10, is about two doublings: a few hours at a two-hour generation time, and less for a young starter (her day-14 1:2:2 and 1:5:5 peaked within half an hour)
- **Temperature buys more.** Below the optimum near 27°C, rates roughly double per 8–10°C; 20°C takes about half again as long
- **Stiffness buys some:** a stiff feed rises slower and holds its peak longer
- **The fridge buys days:** 2–4 weeks unfed

At 24–27°C, peaking in about 6 h on 1:2:2

Away for	Feed
Under time to peak	Current ratio
About 1.5×	One step up
About 2×, workday	1:10:10, or 1:5:5 cooler
Overnight, 12–16 h	1:10:10 cool, or stiff 1:6:10
24–48 h	1:5:5, then the fridge
Days	Fridge

In the kitchen: Ratio buys hours, a cooler spot more, the fridge days.

Source: Full Proof Baking (2019); Gänzle et al. (1998); the Q_{10} rule as in notes/proteolysis-ph-heat.md; chapter 11. The hours are the hypothesis; open test 10 calibrates them

Stirring: harmless to the culture, costly to the measurement

A stir does three things at once:

- **Lets the gas out.** CO_2 is a product, not a resource; the culture loses nothing. The rubber band, time to peak and height at peak read from zero again
- **Moves cells to fresh food.** In a thick batter each colony sits in its own depletion and its own acid; a stir redistributes both **reasoning; not measured in a jar**
- **Folds in air.** Yeast need a little oxygen for membrane sterols, so more cells. *F. sanfranciscensis* turns oxygen into acetate, so sharper. Acetic acid bacteria need it too: a vinegar jar gets worse

Not in the oven

Mixing the hot skin into the core shortens the protective ramp (ch. 10–11), and aerates where *I. orientalis*, the hot team's oxygen-dependent yeast, gains

Before a feed

Pour the liquid off a collapsed jar; do not stir it in. Water, ethanol and acid: a small chapter-11 dilution

In the kitchen: Do not stir a jar you are measuring. Her method stirs once, on day 2.

Source: Andreasen and Stier (1953, 1954); Stolz et al. (1995); Gänzle (2014); Vogelmann and Hertel (2011); Landis et al. (2021); chapters 10–11. The stir-down split test is open test 11

A settled starter: one yeast and a few bacteria

- 500 home starters: a median of one yeast and three bacteria, often one of each dominating
- **Yeast:** wild *S. cerevisiae*, over half the fungal DNA in 77%
- **Bacteria:** most often *L. plantarum* with *L. brevis* (177 of 500)
- *F. sanfranciscensis* dominates where present, mainly in older and bought starters

The bakery classic

F. sanfranciscensis with the yeast *Kazachstania humilis* (ex *Candida milleri*). Two Italian bakeries kept it, with *S. cerevisiae*, stable for three years.

Heat picks the team

The classic pair wins at 20–30°C. At 35–40°C, *L. reuteri* and *L. johnsonii* with the yeast *I. orientalis* win instead.

In the kitchen: A home starter is usually wild *S. cerevisiae* and a couple of lactobacilli. The San Francisco pair is the exception.

Source: Landis et al. (2021); Venturi et al. (2012); Vogelmann and Hertel (2011)

Heat picks the team, and the teams do different jobs

Issatchenkia orientalis is chapters 08–09's *Pichia kudriavzevii*, earlier *Candida krusei*. Type I is a traditional starter that leavens; Type II is a warm, liquid industrial acidifier that gets baker's yeast for lift.

	Cool team, 20–30°C	**Hot team, 35–40°C**
Who	*F. sanfranciscensis* + *K. humilis*; at home *S. cerevisiae* + *L. plantarum*, *L. brevis*	*L. reuteri* + *L. johnsonii*, with *I. orientalis* or no yeast
Leavening	Self-sufficient: baker's yeast's own species, or *K. humilis*, as good in co-culture	Weak: *I. orientalis* grew only with oxygen; Type II doughs get baker's yeast
Acid	Lactic and acetic (ch. 04); acetate falls off at 35°C	Fast and deep, pH toward 3.5 in a day; *L. johnsonii* lactic only, *L. reuteri* some acetic (ch. 09)
Dough	Glutathione reductase in heterofermenters: thiols up, gluten softer and more extensible	Exopolysaccharides from *L. reuteri* strains: softer crumb, more volume
Aroma	Yeast esters and alcohols over the acid	Little from the yeast; glutamate from *L. reuteri* reads as umami

In the kitchen: The cool team leavens, the hot team acidifies. Keep the mother cool.

Source: Vogelmann and Hertel (2011); De Vuyst and Neysens (2005); Brandt et al. (2004); Vermeulen et al. (2006); Tieking and Gänzle (2005); Zhao et al. (2015).

Exopolysaccharides are strain-dependent; the hot team's aroma is thin in the literature

They share the maltose without fighting over it

- The flour's enzymes keep releasing maltose, and sourdough LAB prefer it
- *F. sanfranciscensis* splits maltose; *K. humilis* cannot use it at all, so they don't compete for the main sugar
- Together they use more of it, and leaven as well as LAB paired with *S. cerevisiae*
- "The bacterium leaks glucose to feed the yeast": not universal; Greek sourdough strains showed none
- What kept a bakery pair stable for three years: no main sugar ever ran out

In the kitchen: The partners coexist because the food never runs short, not because they divide it neatly.

Source: Gänzle (2014); Ehrmann and Vogel (1998); Carbonetto et al. (2020); De Vuyst et al. (2002); Venturi et al. (2012)

A one-sided partnership

- **The bacteria gain:** yeasts supply valine and leucine, and lactobacilli grow faster beside them
- *F. sanfranciscensis* grows poorly alone and persists only with *K. humilis*
- **The yeast does not:** its numbers fall when LAB are present; the LAB rarely notice the yeast

Both take the acid; the bacterium more

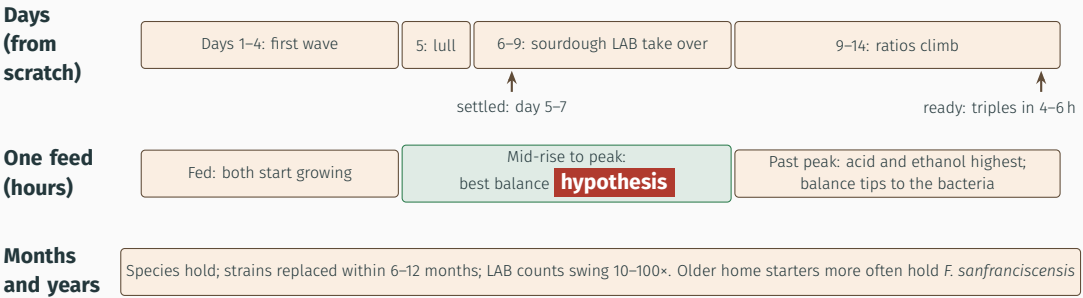
	Grows at	Stops at*
Yeast	pH 3.5–7	150 mmol/L
Bacterium	pH 3.9–6.7	over 160

* undissociated acetic acid; *C. milleri* and *F. sanfranciscensis*

In the kitchen: “Symbiotic” is loose: the bacteria gain from the yeast, the yeast puts up with them, and the pair ferments better than either alone.

Source: Gobbetti et al. (1994); Landis et al. (2021); Carbonetto et al. (2020); Gänzle et al. (1998)

Settled in a week, ready in two, never finished



In the kitchen: Judge readiness by the tripling test, not the calendar, and feed at peak to keep it in its best window.

Source: Van der Meulen et al. (2007); Ercolini et al. (2013); Venturi et al. (2012); Gänzle et al. (1998); Carbonetto et al. (2020); Minervini et al. (2018); Landis et al. (2021)

Video, literature and this kitchen

The method works; three of its explanations do not

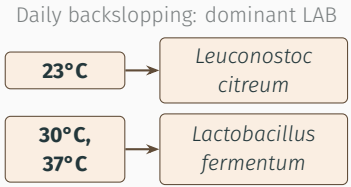
	The video says	The evidence says
1	It “picks up the wild bacteria and yeasts from the air”	**The flour, then the baker’s hands.** Same recipe and flour on two continents: much of each community traced to the flour, the differences to the bakers’ hands. In bakeries over a year, flour drove the LAB; the bakery’s own microbes played a minor role. It happens in a covered jar too
2	Day 3: “We’ve caught some wild yeast!”	In a two-day-old starter the rise is bacterial. Her narration says “not the correct balance” at once: only the label is wrong
3	Day 8: a loose lid, as “this is a live culture that requires air”	Yeast and LAB ferment without oxygen. The lid is loose so CO_2 can escape, not so air can get in

In the kitchen: The cheesecloth does no harm, but it is not what inoculates the jar: the flour is.

Source: Reese et al. (2020); Minervini et al. (2018); Wink (2008a), not peer-reviewed

Start warm, not hot: 24°C or 30°C picks a different culture

- New starters are small and unsettled, like chapter 11's depleted culture: 22–25°C, no selection pressure
- Warm is fine: her target is 27°C; Van der Meulen et al. (2007) had a stable culture in a week at 30°C
- *L. citreum* is behind Wink's false start: at 24°C the first wave may never fully leave
- **hypothesis** begun near 30°C, a starter may suit a warm-kept regime better (open tests 1–2)



Van der Meulen et al. (2011), via chapter 09

In the kitchen: No hot builds until it passes the tripling test; then keep the mother at room temperature and build hot from it.

Source: Van der Meulen et al. (2007, 2011); Wink (2008b), not peer-reviewed; chapters 08, 09, 11

Tap water and ordinary flour will do; juice is optional

Water

- Bottled water is usually about chlorine. Danish tap water is groundwater, normally unchlorinated: tap water should do

Flour

- Any bread flour for the base; the Manitoba is more than strong enough
- Dark wholemeal rye (*fuldkornsrugmel*): the whole grain is the point, as when chapter 11 adds wholemeal in recovery

A shortcut through the lull

- Wink: pineapple juice (or orange juice, or apple cider) for water on days 1–3 acidifies the mix below what the gas producers tolerate
- Yeast on or before day 4, against about day 7 with water
- It “doesn’t change the end result”; never necessary, “only patience”
- Here: juice for the day-1 mix and the day-3 feed **not yet tested**

Source: Wink (2008a, b), not peer-reviewed. Juice in this method is the note’s suggestion and has not been tried.

Drying a backup and reviving it

A pinch worked where a soaked spoonful failed own result

First attempt

A recipe worked out in a September 2026 conversation with Claude:

- a larger amount of flakes
- soaked in warm water first

It did not take.

October 2026

Flakes a year old:

- a good pinch, about 1 g, crushed into a fresh 30 g water + 30 g wholemeal/rye mix
- no separate soak

Clearly active in several days.

Why keep a backup: after a heat kill, it is the only way back to the original culture, rather than to whatever the flour supplies. That holds for the bacteria; the yeast may not come through drying.

Source: Own runs, September and October 2026; chapter 11, *Recovery Protocol*

Dry it cool: a wet film is heat-fragile, a dry flake is not

Drying it

- Spread thin on baking paper; room temperature until it snaps, about 24–48 h
- Crumble; store airtight, dry, cool and dark
- No oven, no warm dehydrator: the guide allows one only below 37°C

Dry is heat-proof; drying is not

- Until dry, the film is a wet culture: six hours or more at 40–45°C is high risk, and the D-value is under a minute at 55°C (chapter 11)
- Dried *S. cerevisiae* kept full viability through 60 min at 75°C, and 10 min at 100°C at 12% relative humidity
- The September chat said dried cells lose heat tolerance: backwards

In the kitchen: The heat risk is the wet start of drying, not the dry end: dry at room temperature.

Source: Country Roads Sourdough, not peer-reviewed; Ribert et al. (2021); chapter 11

Reviving it in five steps, with amounts by eye own result

1. **Start:** crush a good pinch of flakes, about 1 g, into about 30 g water + 30 g flour, half wholemeal, half rye. Water at room temperature or a little above, not cold; no separate soak
2. **Wait, unfed, for several days** at 24–26° C until clearly active. A dry crust forms on top: the day-2 skin, harmless
3. **Lift off the crust and feed what is underneath** with bread flour and wholemeal at about 1:1:1. From the next feed, keep 20 g and discard the rest, as on day 4
4. **Feed at peak** as in the main method; one or two feeds on the usual flour before baking with it
5. **Dry a new backup** once it doubles reliably

In the kitchen: Record flake age, rough amount, water temperature, first sour smell, first rise, first doubling. Sour without a rise: the LAB are back, the yeast not yet.

Source: The method as run here, October 2026; amounts estimated, not weighed, and it does not seem sensitive to them

A dried starter is not a paused starter

What	Through drying	Why
Water	Leaves, to equilibrium with the room	—
CO_2 and ethanol	Leave	A gas and a volatile, from a thin film
Acetic acid	About half leaves	Volatile: 57% kept in freeze-drying
Lactic acid	Mostly stays	Not volatile: 85% kept in freeze-drying
Spent flour: starch, gluten, bran, minerals	Stays	Already part-fermented: little food left in it
Flour enzymes	Stay, idle until wet	Flour is itself a dry product whose enzymes work once wetted
Cells	A minority survive dormant; most die	Next slide

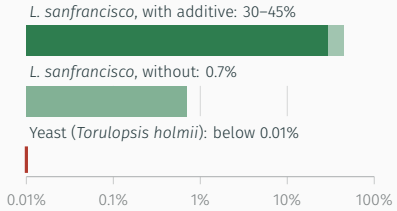
In the kitchen: Slow air-drying probably loses at least as much acetic acid, so a flake brings back mostly lactic acid. That is why the dose matters.

Source: Acid figures: Kline (1979), a patent, not peer-reviewed

Drying kills most cells, the yeast hardest

- First hours: the film keeps fermenting as sugars, salts and acids concentrate
  reasoning; nothing measured
- Damage: mechanical, structural and oxidative in yeast; in LAB mainly the membrane, as it changes phase
- Air-drying *S. cerevisiae*: under 1 in a million growing cells survive; 20–40% of cells that ran out of food
- **hypothesis** so dry later than the usual “at peak”? Uncertain: tolerance comes with respiration, a jar is short of oxygen, no sourdough yeast was tested (open test 4)

Survival, freeze-dried San Francisco starter



Kline (1979) **a patent, not peer-reviewed**

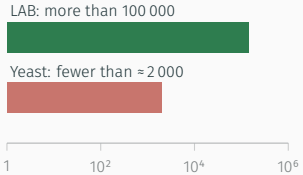
Freeze-drying is not air-drying, but this is the only sourdough-specific comparison found.

Source: Calahan et al. (2011); Dupont et al. (2014); Santivarangkna et al. (2008); Kline (1979), a patent, not peer-reviewed

The bacteria come back on numbers, not toughness

- Not toughness: without the additive, over 99% of Kline's bacterium died
- Numbers: far more LAB than yeast (≈ 10^7/g) to start, and a smaller share lost
- Why the bacterium dies less is not established (Potts 1994)
- Not the acid: acid-adapted *L. lactis* dried no better; stopping growth is shown to protect only yeast
- Flour may protect a little **not yet tested**

Survivors in a 1 g pinch, before storage



borrowed figures, not a count

In the kitchen: Sour comes back first because the bacteria were never scarce; the rise waits on the yeast.

Source: Kline (1979), a patent, not peer-reviewed; Van der Meulen et al. (2007); Potts (1994); Calahan et al. (2011); Cnossen et al. (2020)

Water starts four things at once

Rehydration

1. The lactic acid redissolves. A pinch gives the fresh flour a small dose; a spoonful gives close to full-strength starter acid

2. Flour enzymes start freeing sugar again, in the fresh flour and any left in the flake

3. Survivors take up water and their membranes pass back through the phase change: the moment of leakage, worse in cold water

4. Dead cells leak their contents into the mix

Then a lag, while the survivors repair before they divide. From there the revival is a race between their regrowth and the flour's own microbes.

In the kitchen: Step 3 is where water temperature matters: the leakage is worse in cold water.

Source: Santivarangkna et al. (2008); Crowe et al. (1989)

A pinch skips the first wave; the yeast still has to build up

From a pinch of dried flakes own result

Start: about 1 g crushed flakes + 30 g water + 30 g wholemeal/rye

First hours: lactic acid redissolves · flour enzymes free sugar · survivors rehydrate and repair before dividing

Surviving LAB regrow: many survivors, already acid-tolerant · pH falls below 4.5 → the flour's gassy first wave is shut out: no false rise, no lull

Yeast: perhaps a hundred survivors, maybe none, + newcomers from flour, hands, kitchen · the acid admits the tolerant ones

Several days, unfed: sour but no rise · dry crust on top · yeast regrows over about 20 doublings

First rise: lift off the crust · feed bread flour + wholemeal 1:1:1 · then keep 20 g, as on day 4

Feed at peak until it doubles reliably → dry a new backup

From scratch, for comparison

Days 1–4: the flour's first wave, *Leuconostoc*, *Enterobacter* · big rise, bad smell

skipped

Day 5: the lull · acid stops the first wave

Days 6–9: sourdough LAB take over · yeast the day after pH 3.5

Days 9–14: ratios climb until it triples

Source: own run, October 2026, for the timings. The order of events inside them is reasoning **no pH or counts were taken**

The surviving bacteria close the window the first wave needs

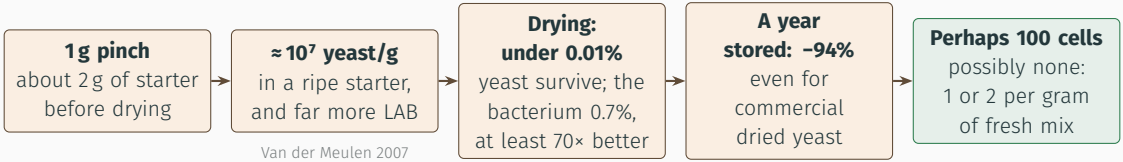
- From scratch, a gassy first wave rises and collapses and the jar goes quiet before the sourdough organisms take over
- The revival took days, but had **no false start and no lull** **own result**
- The likely reason is the inoculum: the surviving LAB are sourdough organisms, selected by months of feeding, that grow well at their own acidity
- They acidify the fresh flour before its first wave builds, closing Wink's window: the gas producers grow at pH ≥ 5 and not at ≤ 4.5
- From scratch nothing acid-tolerant is there in numbers, so the first wave gets days

reasoning that fits the run here; no pH was measured

In the kitchen: From flakes, expect quiet days that turn sour before rising: no day-4 surge, no lull.

Source: Wink (2008a), not peer-reviewed; own run, October 2026

The yeast restarts from perhaps a hundred cells, possibly none



- That is over 20 doublings short of a ripe starter: each tenfold is about 3.3 doublings
- The LAB start from far more survivors. The revival behaved as if the LAB had come through and the yeast had to be caught again **own result**
- **arithmetic on borrowed figures, not a count**: Kline's starter was freeze-dried, and Ribert's yeast was grown to be dried

Source: Van der Meulen et al. (2007); Kline (1979), a patent, not peer-reviewed; Ribert et al. (2021)

It skipped the bacterial half of the first week, not the yeast half

- **The yeast may not be the original.** Survivors compete with yeast arriving as it does from scratch: with the flour, and from the baker's hands
- A kitchen that kept the starter for years may hold its yeast: in four bakeries, the sourdough's dominant yeast also dominated walls, air, storage boxes and mixers. A "new" yeast may be the old one coming back **loose analogue**
- The yeast does not arrive in a from-scratch jar: the LAB have already acidified it. From scratch, yeast appeared the day after pH 3.5, so its clock starts at once, and the acid favours tolerant yeasts

In the kitchen: A backup reliably brings back the bacteria, and perhaps the yeast. Drying later may save more of it (open test 4); whether it is the old yeast is open test 7.

Source: Reese et al. (2020); Minervini et al. (2015): bakeries hold far more flour and dough than a home, and the study cannot say which way the yeast moved; Wink (2008a), not peer-reviewed

A pinch, not a spoonful: the flakes carry the old starter's acid

A spoonful
in a little water
and flour

↓

Close to full-strength starter acid, little fresh food: chapter 11's dying culture

first attempt, also warm-soaked: did not take **own result**

A pinch
in 60 g fresh wholemeal and rye

↓

Far more dilute, plenty of food; wholegrain is what chapter 11 adds

worked **own result**

- Flakes are the old flour plus all its lactic acid, which does not evaporate in drying
- Chapter 11 dilutes a dying culture out at 1:10:10
- Country Roads' recipe is the high-dose kind, 5 g of flakes into 20 g water + 20 g flour, and expects about a week
- Two things changed, amount and soak: one run cannot separate them (open test 6). The soak is the less likely culprit: 35–38°C for half an hour is brief by chapter 11's standards, unless the water was hotter

Source: Own runs, 2026; chapter 11, *Recovery Protocol*; Country Roads Sourdough, not peer-reviewed

Water: not cold. The documented hazard is cold, not warm

- The September chat advised lukewarm, 25–30°C, because cells are most fragile as water rushes back in. The fragility is real; the direction is wrong
- As a dry cell takes up water, its membrane lipids pass from gel to fluid. In cold water the membrane leaks while it changes state: **imbibitional damage**, measured in pollen, with dry yeast named as a case
- Active dry yeast rehydrated at 0°C against 40°C: less respiration and fermentation, membranes that did not reseal even after hours at 40°C, few cells growing on plates
- Limits: pollen; pure commercial *S. cerevisiae*; flakes in flour may wet more slowly; nothing about 20°C. The direction is supported; the room-temperature penalty is unknown (test 5)

In the kitchen: Mix with water at room temperature or a little above, up to about 35°C, then hold the jar at 24–26°C. Avoid cold tap water.

Source: Crowe et al. (1989); Peña et al. (1992); Country Roads Sourdough (room-temperature water; cold only slows things), not peer-reviewed

A backup loses ground every month, but slowly

Source	What was stored	Loss
Calahan et al. (2011)	Air-dried *S. cerevisiae*	About 20% survival over a month
Ribert et al. (2021)	Dried *S. cerevisiae*, 25°C, 55% relative humidity	1.24 log, about 94%, in a year
Hajar-Azhari et al. (2024)	Spray-dried, kombucha-started sourdough powder, best of four packagings **loose analogue**	About 2.5 log of LAB and yeast in 90 days
Here **own result**	Flakes a year old (October 2026)	Back from a pinch in several days

- A dry flake follows the room: water activity about 0.5 in a 50% kitchen, far below anything that grows. Losses are mainly oxidation of membrane lipids: hence airtight, cool, dark
- Each tenfold loss costs about 3.3 doublings: an old backup wakes more slowly, not never. True of the LAB; the yeast is less certain

In the kitchen: Replace the backup once a year from the live starter to keep the loss small.

Source: As in the table; FDA (1984); Santivarangkna et al. (2008); Country Roads Sourdough, not peer-reviewed; chapter 11

Takeaways

- The day-4 rise is bacteria and the day-5 lull is normal: keep to the schedule through both
- Feed at peak, and set the ratio by how often you feed: 1:2:2 is for three feeds a day; held twice a day, it fades
- Done means tripling in 4–6 h on 1:2:2 at 27°C. No hot builds before that
- Start warm, not hot: 24°C or 30°C is a choice of culture as well as pace
- The microbes come from the flour and your hands, not the air
- Dry a backup thin at room temperature; store it airtight, cool and dark; replace it yearly
- Revive with a pinch in a fresh wholemeal and rye mix, water not cold. The bacteria come back first; the yeast later, and perhaps a new one
- Away longer than its time to peak? Ratio buys hours, a cooler spot more, the fridge days

Open tests

Open tests 1–7: from scratch, and the dried backup not yet tested

Starting from scratch

1. **Start temperature.** Two jars, same flour and day-1 mix: room temperature against 30°C (the chamber can hold it). Compare the end of the lull, the first tripling at 1:2:2, the smell. Van der Meulen et al. (2011) predicts different dominant LAB
2. **Does a 30°C start take to the hot protocol better** than a 24°C one? Only once both pass the tripling test, on split lineages (chapter 08)
3. **Juice versus water**, Wink's own design: two identical jars, one with juice on days 1–3

The dried backup

4. **Dry at peak or after it.** Two films from one jar, at peak and a few hours after the fall; revive equal weights; compare the first doubling. Calahan et al. (2011) predicts the later film wins
5. **Water temperature.** Same flakes, 35°C water against cold tap, both then at 24–26°C. Peña et al. (1992) predicts the cold jar is slower
6. **Pinch versus spoonful**: a pinch against 10 g, same flour and water temperature. The acid account predicts a slower spoonful; a pH strip at the start should show why
7. **Same yeast or a new one?** No kitchen test; sequencing the fungal ITS region of flakes and revived starter, as Reese et al. (2020) did, would. Dead cells keep their DNA; a species match would not prove the same strain

Source: The Open tests section of notes/starter-from-scratch.md

Open tests 8–11: keeping it strong not yet tested

8. **Does a higher ratio help the revived yeast catch up?** Split the revived starter, 1:2:2 against 1:5:5 at the same temperature, each fed at its own peak. After two weeks feed both 1:2:2 side by side and compare time to peak and height. The dilution account predicts the 1:5:5 jar is faster
9. **Does the slide reverse with dilution?** From 2026-10-31: the from-scratch starter that faded under 1:2:2 to 1:3:3, now on 1:5:5, and a jar left underfed in the fridge, on chapter 11's 1:10:10; both with wholemeal at 24–27°C, fed at peak. Record feeds to the first 2.5× peak and to the tripling test. Chapter 11 predicts days, not weeks; the slide account predicts no new organisms are needed, since only the yeast's share fell
10. **Hours per step, and per degree.** One mother, three jars at 1:2:2, 1:5:5 and 1:10:10 at one measured temperature; time to peak and height for each. Then one ratio at 20°C against 27°C. The two numbers calibrate the planning table. Prediction: about one generation per doubling of the dilution, and almost no difference while the starter is young
11. **What a stir-down does.** One feed into two jars; stir one down at half height. Compare the stirred jar's second peak, height and time from the stir, against the control's single peak, and smell both. Redistribution predicts a higher second peak; oxygen predicts a sharper smell; a film or vinegar within a few feeds is item 8's warning

In the kitchen: All four read off time to peak and height at peak, from a split of one feed.

Source: The Open tests section of notes/starter-from-scratch.md; chapter 11, *Recovery Protocol*

Sources

The method and the first two weeks

- **Kristen, Full Proof Baking (2019)** *How To Make A Sourdough Starter From Scratch.* YouTube, 7 Feb 2019.
- **Wink D (2008a)** The pineapple juice solution, Part 1. *Bread Lines* 16(1). Not peer-reviewed.
- **Wink D (2008b)** The pineapple juice solution, Part 2. *Bread Lines*, June 2008. Not peer-reviewed.
- **Van der Meulen R et al. (2007)** *Appl. Environ. Microbiol.* 73(15):4741–4750.
- **Van der Meulen R et al. (2011)** *Appl. Environ. Microbiol.* 77(8):2716–2726.
- **Ercolini D et al. (2013)** Microbial ecology dynamics during rye and wheat sourdough preparation. *Appl. Environ. Microbiol.* 79(24):7827–7836.
- **Minervini F et al. (2018)** LAB population dynamics over one year of daily propagations. *Frontiers in Microbiology* 9:1984.
- **Reese AT et al. (2020)** Influences of ingredients and bakers on the bacteria and fungi in sourdough starters and bread. *mSphere* 5(1):e00950-19.
- **Minervini F et al. (2015)** House microbiotas as sources of LAB and yeasts in traditional Italian sourdoughs. *Food Microbiology* 52:66–76.

Sources (cont.)

The finished culture

- **Landis EA et al. (2021)** The diversity and function of sourdough starter microbiomes. *eLife* 10:e61644.
- **Venturi M et al. (2012)** Stable and non-competitive association of *S. cerevisiae*, *C. milleri* and *L. sanfranciscensis*. *Food Microbiology* 31(1):107–115.
- **Carbonetto B et al. (2020)** Interactions between *K. humilis* and lactic acid bacteria in sourdough. *Microorganisms* 8(2):240.
- **Gobbetti M et al. (1994)** Interactions between LAB and yeasts: metabolism of amino acids. *World J. Microbiol. Biotechnol.* 10(3):275–279.
- **Vogelmann SA, Hertel C (2011)** Ecological factors and the stability of sourdough associations. *Food Microbiology* 28(3):583–589.
- **Ehrmann MA, Vogel RF (1998)** Maltose metabolism of *L. sanfranciscensis*. *FEMS Microbiol. Lett.* 169(1):81–86.
- **De Vuyst L et al. (2002)** LAB in Greek traditional wheat sourdoughs. *Appl. Environ. Microbiol.* 68(12):6059–6069.
- **Gänzle MG (2014)** Enzymatic and bacterial conversions during sourdough fermentation. *Food Microbiology* 37:2–10.
- **Gänzle MG et al. (1998)** Modeling of growth of *L. sanfranciscensis* and *C. milleri*. *Appl. Environ. Microbiol.* 64(7):2616–2623.

Sources (cont.)

- **De Vuyst L, Neysens P (2005)** The sourdough microflora: biodiversity and metabolic interactions. *Trends Food Sci. Technol.* 16(1–3):43–56.
- **Brandt MJ, Hammes WP, Gänzle MG (2004)** Effects of process parameters on growth and metabolism of *L. sanfranciscensis* and *C. humilis* during rye sourdough fermentation. *Eur. Food Res. Technol.* 218(4):333–338.
- **Vermeulen N et al. (2006)** Influence of redox-reactions catalysed by homo- and hetero-fermentative lactobacilli on gluten in wheat sourdoughs. *J. Cereal Sci.* 43(2):137–143.
- **Tieking M, Gänzle MG (2005)** Exopolysaccharides from cereal-associated lactobacilli. *Trends Food Sci. Technol.* 16(1–3):79–84.
- **Zhao CJ et al. (2015)** Glutamate accumulation during sourdough fermentation with *L. reuteri* and the taste of bread. *Cereal Chemistry* 92(2):224–230.
- **Andreasen AA, Stier TJB (1953, 1954)** Anaerobic nutrition of *S. cerevisiae*: I. ergosterol and II. unsaturated fatty acid requirements. *J. Cell. Comp. Physiol.* 41(1):23–36 and 43(3):271–281.
- **Stolz P, Böcker G, Hammes WP, Vogel RF (1995)** Utilization of electron acceptors by lactobacilli isolated from sourdough. I. *L. sanfrancisco*. *Z. Lebensm. Unters. Forsch.* 201(1):91–96.

Drying and reviving

Sources (cont.)

- **Country Roads Sourdough** *Dehydrating and Rehydrating Sourdough Starter: A How To Guide.* Not peer-reviewed.
- **Kline L (1979)** Freeze-dried natural sour dough starter. US Patent 4,140,800. A patent, not peer-reviewed.
- **Calahan D et al. (2011)** Genetic analysis of desiccation tolerance in *S. cerevisiae*. *Genetics* 189(2):507–519.
- **Dupont S et al. (2014)** Survival kit of *S. cerevisiae* for anhydrobiosis. *Appl. Microbiol. Biotechnol.* 98(21):8821–8834.
- **Santivarangkna C et al. (2008)** Inactivation mechanisms of LAB starter cultures preserved by drying. *J. Appl. Microbiol.* 105(1):1–13.
- **Crowe JH et al. (1989)** Membrane phase transitions are responsible for imbibitional damage in dry pollen. *PNAS* 86(2):520–523.
- **Peña A et al. (1992)** Rehydration temperature is critical for metabolic competence and membrane integrity in active dry yeast. *Arch. Microbiol.* 158(2):75–80.
- **Ribert P et al. (2021)** Effect of devitrification on the survival and resistance of dried *S. cerevisiae*. *Appl. Microbiol. Biotechnol.* 105(16–17):6409–6418.
- **Hajar-Azhari S et al. (2024)** Sourdough powder: shelf-life stability in different package type. *J. Food Sci. Technol.* 61(9):1701–1710.
- **Potts M (1994)** Desiccation tolerance of prokaryotes. *Microbiol. Rev.* 58(4):755–805.

Sources (cont.)

- **Cnossen DC et al. (2020)** Adaptive stresses and the survival of spray-dried *Lactococcus lactis*. *J. Dairy Res.* 87(3):382–385.
- **US FDA (1984)** *Water Activity (aw) in Foods*. Inspection Technical Guide.

Full entries: the References section of notes/starter-from-scratch.md, and chapter 11.